

An Extension of OEIS A255885 by an Interval-Aware Inverse Root Sieve

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Abstract

For an integer base $b \geq 2$, let

$$C(b) = \#\{c : 2 \leq c < b, \text{ } c \text{ composite, } b^{c-1} \equiv 1 \pmod{c^2}\},$$

and let $a(n)$ be the least b for which $C(b) = n$. This is OEIS A255885. We replace a direct base-by-base search by an inverse sieve over composite moduli. For each c the roots of $x^{c-1} = 1 \pmod{c^2}$ are constructed from prime-power roots and the Chinese remainder theorem. When c^2 exceeds the base bound, an interval-aware projection enumerates roots modulo a suitably chosen divisor square and certifies the omitted local conditions. A production computation through $b = 2,000,000,000$ and an independent arithmetic recomputation establish 61 new consecutive terms, $a(25)$ through $a(85)$, together with 23 exact nonconsecutive values. Among the 254 inspected indices, 108 are therefore realized (including the 24 previously known terms), while the remaining 146 have no realizing base within the computed frontier. In particular,

$$a(85) = 1,244,678,401, \quad a(86) > 2,000,000,000.$$

The independent validator compared all 2,000,000,001 counter cells, all 254 result slots (minimum or undefined), and all 5,725 witnesses belonging to the 84 newly determined indices.

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1 Introduction

For each $b \geq 2$, define

$$C(b) = \#\{c \in \mathbb{Z} : 2 \leq c < b, \text{ } c \text{ composite, } b^{c-1} \equiv 1 \pmod{c^2}\}. \quad (1)$$

OEIS A255885 records

$$a(n) = \min\{b \geq 2 : C(b) = n\}. \quad (2)$$

The index starts at $n = 1$. Programs that state $c \leq b$ instead of $c < b$ are equivalent here: if $c = b$ is composite, then b^{b-1} is divisible by b^2 and therefore cannot be congruent to 1 modulo b^2 . The public entry, created by Felix Fröhlich in 2015 and extended by Chai Wah Wu in 2022, contained the 24 values through $a(24) = 389377$ when the present computation was undertaken [3]. The sequence is not monotone: for example, the new values include $a(28) = 902501$ but $a(29) = 19601$.

The congruence in (1) is expensive to test in the forward direction for every pair (b, c) . The decisive change of viewpoint is to fix c , solve

$$x^{c-1} \equiv 1 \pmod{c^2}, \quad (3)$$

and increment precisely the bases represented by those roots. Robert Israel's program for OEIS A273785 already used this inverse-modulus principle by solving the roots modulo c^2 and marking their progressions [4]. The present work adds an interval-aware projection that avoids constructing the full root set modulo c^2 when c^2 is larger than the finite base interval.

Agoh, Dilcher, and Skula studied the related Euler–Wieferich congruence $b^{\varphi(c)} \equiv 1 \pmod{c^2}$ and its factorwise p -adic structure [1]. Since $c-1$ and $\varphi(c)$ are different exponents, that congruence does not define A255885. It is, however, a necessary consequence of every incidence counted here and supplies a redundant structural consistency check. Indeed, the defining congruence forces every principal-unit component to be trivial, while the remaining root order divides the relevant factors $p-1$ and hence divides $\varphi(c)$.

This paper gives the root decomposition, proves the completeness of the interval-aware sieve, describes the production and validation paths, and reports the exact finite results. The computation was bounded in advance at $2 \cdot 10^9$; no claim is made about the value of $a(86)$ beyond the proved lower bound.

2 Prime-power roots

Write

$$c = \prod_{i=1}^s p_i^{e_i}, \quad k = c - 1.$$

The primes are ordered $p_1 < \dots < p_s$, and $p^e \parallel c$ means that p^e divides c but p^{e+1} does not. We use standard facts about finite unit groups, primitive roots, and the Chinese remainder theorem as collected, for example, by Cohen [2]. Because $\gcd(k, c) = 1$, in particular $\gcd(k, p_i) = 1$ for every prime divisor p_i of c .

Lemma 1 (Local root count). *For an odd prime power $p^e \parallel c$, the congruence*

$$x^k \equiv 1 \pmod{p^{2e}}$$

has exactly $\gcd(k, p-1)$ solutions. For $2^e \parallel c$, it has exactly one solution, namely $x \equiv 1 \pmod{2^{2e}}$.

Proof. If $p \mid x$, then $x^k \equiv 0 \pmod{p}$, so every solution is a unit. For odd p , the unit group modulo p^{2e} is cyclic of order $p^{2e-1}(p-1)$. The kernel of exponentiation by k therefore has size

$$\gcd(k, p^{2e-1}(p-1)) = \gcd(k, p-1),$$

because $p \nmid k$. If $p = 2$, then c is even and $k = c - 1$ is odd. Exponentiation by an odd integer is an automorphism of the finite 2-group $(\mathbb{Z}/2^{2e}\mathbb{Z})^\times$ (equivalently, of $C_2 \times C_{2^{2e-2}}$ when $e \geq 1$), so its kernel is trivial. $\square$

Corollary 2 (Global root count). *The number of roots of (3) modulo c^2 is*

$$K(c) = \prod_{\substack{p \mid c \\ p \text{ odd}}} \gcd(c-1, p-1). \quad (4)$$

All roots are obtained uniquely by combining the local roots with the Chinese remainder theorem.

The product in (4) cannot in general be replaced by $\gcd(c-1, \varphi(c))$. For instance, $c = 15$ has four roots modulo 225, whereas that gcd is two. This small example is also a useful implementation test.

3 The interval-aware inverse sieve

Fix a finite base bound B . If $c^2 \leq B$, the algorithm enumerates every root $r \pmod{c^2}$ and every integer

$$b = r + tc^2 \quad \text{with} \quad c < b \leq B.$$

The difficulty is the much larger set of moduli satisfying $c^2 > B$. Let $P(c) = \{p_1^{e_1}, \dots, p_s^{e_s}\}$. For a subset $S \subseteq P(c)$ set

$$d(S) = \prod_{p_i^{e_i} \in S} p_i^{e_i}, \quad K_c(S) = \prod_{\substack{p_i^{e_i} \in S \\ p_i \text{ odd}}} \gcd(c-1, p_i-1).$$

Thus $K(c) = K_c(P(c))$. A subset is admissible when $d(S)^2 > B$; at least one exists because the full set gives $d(S) = c$. The production code chooses an admissible subset that minimizes $K_c(S)$ and, on a tie, takes the smallest numeric bit mask in the increasing-prime order. This choice affects work only: every admissible subset gives the same accepted bases. Roots are first combined only modulo $d(S)^2$. Since $1 \leq b \leq B < d(S)^2$, each retained residue represents at most one base in the search interval.

The prime-power factors omitted from $d(S)$ are then checked without constructing their roots. The following criterion is used in the production code.

Proposition 3 (Exact omitted-factor tests). *Let $p^e \parallel c$ be omitted from $d(S)$ and let $k = c - 1$.*

1. *If p is odd, then*

$$b^k \equiv 1 \pmod{p^{2e}}$$

is equivalent to the conjunction

$$b^{\gcd(k, p-1)} \equiv 1 \pmod{p}, \quad b^{p-1} \equiv 1 \pmod{p^{2e}}. \tag{5}$$

2. *If $p = 2$, then*

$$b^k \equiv 1 \pmod{2^{2e}} \iff b \equiv 1 \pmod{2^{2e}}.$$

Proof. For odd p , if $p \mid b$, then both the displayed congruence on the left and the first congruence in (5) fail. We may therefore assume $p \nmid b$. Let $h = \gcd(k, p-1)$. Write the unit group as

$$(\mathbb{Z}/p^{2e}\mathbb{Z})^\times \simeq \mu_{p-1} \times U_1, \quad |U_1| = p^{2e-1},$$

and write $b = \omega u$ in its Teichmüller and principal-unit components. Since $p \nmid k$, exponentiation by k is an automorphism of U_1 ; hence $b^k = 1$ is equivalent to $u = 1$ and $\omega^k = 1$. The first congruence in (5) is equivalent to $\omega^h = 1$, hence to $\omega^k = 1$. The second is equivalent to $u^{p-1} = 1$, hence to $u = 1$ because $\gcd(p-1, |U_1|) = 1$. This also proves the converse. For $p = 2$, Lemma 1 gives the unique root. $\square$

Theorem 4 (Completeness and uniqueness). *For every composite $c < B$, the interval-aware inverse sieve emits each b satisfying $c < b \leq B$ and $b^{c-1} \equiv 1 \pmod{c^2}$ exactly once and emits no other base.*

Proof. When $c^2 \leq B$, Corollary 2 gives every root modulo c^2 , and distinct root–progression pairs give distinct bases. Suppose $c^2 > B$. Any valid b reduces to a projected root modulo $d(S)^2$; because $b \leq B < d(S)^2$, that residue identifies b uniquely. The projected branch retains the representative only when $c < b \leq B$. Proposition 3 accepts precisely when every omitted prime-power congruence also holds. The Chinese remainder theorem then shows that the accepted b is a full root modulo c^2 . Conversely, every full root passes all projected and omitted conditions. The factorization into complete prime-power factors and uniqueness of the CRT representation prevent duplication. $\square$

4 Implementation and finite coverage

The production implementation is written in C17 with OpenMP. A smallest prime factor table supplies the factorization of every $c < B$ and identifies compositeness by table lookup, with no probabilistic primality test. For an odd p , the production code constructs a primitive root modulo p from the factorization of $p - 1$; if necessary it replaces g by $g + p$ to obtain a primitive root modulo p^2 , which then remains primitive at higher powers. The local kernel is generated by $g^{\varphi(p^{2e})/\gcd(k, p-1)}$; the unique local root at $p = 2$ is 1. Recursive CRT combination produces the projected or full roots.

The numeric domain is explicit. The bound, moduli c , and smallest-prime-factor entries use 32-bit unsigned integers. Squares and local moduli use 64-bit unsigned integers. Products in modular multiplication and CRT use `unsigned __int128` intermediates, followed by checked conversion; compile-time assertions and run-time checks reject any admitted bound or intermediate that would exceed these domains.

In outline, the production algorithm is:

1. build a 32-bit smallest-prime-factor array through $B - 1$;
2. process each composite c exactly once with dynamic OpenMP scheduling in chunks of 64, factor c , and choose the projection subset;
3. enumerate local roots and combine them by CRT;
4. if $c^2 \leq B$, mark all progressions with $c < b \leq B$; otherwise use the unique projected representative, apply the same range test, and certify every omitted factor with Proposition 3;
5. merge the private counter arrays and extract the first occurrence of each requested count.

This is an output-sensitive exhaustive sieve: its arithmetic work is governed by the enumerated projected roots and accepted progression points, while its dominant live allocation is stated exactly below.

Each thread owns a `uint8_t` array indexed from 0 through B and increments with a clamp at 255. After the OpenMP join, arrays are merged in increasing thread order: both byte values are promoted to an unsigned type, added, and clamped again at 255. The per-thread arrays are first merged into the current segment delta; that delta is then combined with the active checkpoint by the same operation. Although the official run used one computational segment, this is the general persistence path. Saturating addition is associative and commutative for the scientific counts, while the fixed merge order also makes the file-level result reproducible. Thus the stored cell is $\min(C(b), 255)$. Because the requested indices are $1 \leq n \leq 254$, saturation preserves every relevant equality. The completed run recorded zero saturation events, and the largest observed count was 160, so the cap was inactive in this campaign.

The final bound was selected before the official run from the reviewed numeric and operational envelope, not from a pattern in the resulting terms. The host had 64 GB DDR5 installed RAM. Immediately before the production invocation, Linux reported 59,471,964 KiB, or about 56.7 GiB, as available. Here 1 KiB = 2^{10} bytes and 1 GiB = 2^{30} bytes. The operating-system measurement and the allocation projections are stated in binary units because they derive from binary byte counts. After complete calibrations at 10^8 and $5 \cdot 10^8$, the dominant live-allocation formula was

$$(6 + T)(B + 1) \text{ bytes} + 64 \text{ MiB},$$

where T is the production thread count. The coefficient 6 comprises the 4-byte smallest-prime-factor entry, the 1-byte active checkpoint count, and the 1-byte current segment delta for each index; the T term comprises the private 1-byte counter arrays, and the final 64 MiB is bounded scratch and overhead allowance. Thus the initial 13–16-thread projections at $B = 2 \cdot 10^9$ ranged from about 35.45 to 41.04 GiB. The final five-thread configuration projected 20.55 GiB, consistent with the measured peak RSS of about 20.49 GiB. At $B = 4 \cdot 10^9$, the same 16-thread layout would project about 82.02 GiB and exceed the host envelope. A five-thread layout would project about 41.04 GiB and might fit in memory, but that frontier had not been calibrated, reviewed, or shown to have a practical elapsed time. Accordingly, $B = 2 \cdot 10^9$ was the largest prevalidated frontier with a comfortable resource margin; the limit was operational rather than a mathematical barrier or a claim that no redesign could go farther.

The run used a single exhaustive modulus segment $4 \leq c \leq 1,999,999,999$ and covered every base $2 \leq b \leq 2,000,000,000$; the allocated counter cells at indices 0 and 1 were unused. It processed 1,901,777,711 composite moduli, generated 273,026,083,586 projected roots, and performed 628,611,669 counter increments. The run environment was 64-bit Linux on an AMD Ryzen 9 7940HS processor running Linux Mint 22.3, with code compiled by GCC 13.3.0 and the GNU OpenMP runtime. The release flags were

```
-std=c17 -Wall -Wextra -Wpedantic -Wshadow -Wconversion
-Wdouble-promotion -Wformat=2 -Werror -fopenmp -O3 -flto
-funroll-loops -fomit-frame-pointer -march=native -mtune=native
```

The main production invocation took about 916 seconds wall time using five OpenMP threads; its kernel phase took about 734 seconds and peak resident memory was 21,486,820 KiB. Initialization, computation, status, and internal verification together took about 1,231 seconds; the 916-second production invocation is included in this total. These figures are execution measurements, not asymptotic performance claims.

The production verification path fails closed unless the artifact schema and checksums are valid, the embedded source and build identities agree with the configuration, manifest continuity and the completed frontier are intact, the active checkpoint hash and slot agree, and a regenerated result table is semantically identical to the stored one. These are persistence and derivation checks; they deliberately do not substitute for the independent arithmetic certificate described next.

5 Independent recomputation and certification

The independent validator has a separate C17 arithmetic engine and a Python certificate layer. Its C engine builds its own smallest-prime-factor table, uses shared exact 32-bit atomic counters instead of thread-local saturating bytes, constructs odd local roots by Hensel lifting rather than by the production generator, breaks projection ties by larger d^2 and then larger bit mask rather than by the production rule, and checks omitted factors directly with exponent $c - 1$. It then

compares every counter cell and every result slot against the production result. The Python layer independently factors each witness modulus, recomputes the defining congruence, and verifies the necessary but logically redundant Euler–Wieferich consistency condition

$$b^{\varphi(c)} \equiv 1 \pmod{c^2}. \quad (6)$$

The final validation recomputed the complete array of 2,000,000,001 counters. It compared all 254 minimum-or-undefined rows, finding zero disagreements in either comparison, and checked all 5,725 witness rows for the 84 new targets. Both implementations also reproduced the 24 pre-existing OEIS terms exactly. The figure 5,725 is also the sum of those 84 newly determined indices, since a result at index n has exactly n witnesses.

The independent arithmetic engine used 16 OpenMP threads, took about 426 seconds wall time (about 403 seconds in its arithmetic phase), and reached 17,582,416 KiB peak resident memory. Its shorter elapsed time than the five-thread production run is attributable principally to the larger thread team, not evidence that its arithmetic backend is intrinsically faster. The complete validator runner took about 436 seconds. Every comparison passed.

This is independent implementation evidence, not a claim of logically unrelated mathematics: both programs necessarily use the same prime-power decomposition, CRT framework, and inverse-modulus identity. The different root constructors, counter representations, projection choices, and direct final congruence checks reduce implementation common-mode risk, while the mathematical proofs above carry the completeness claim.

6 Exact results

For integer indices, $[u, v]$ denotes the set $\{u, u+1, \dots, v\}$.

Theorem 5 (Certified finite extension). *For the function in (2), the values $a(25)$ through $a(85)$ are those in Table 1. There is no base $b \leq 2,000,000,000$ with $C(b) = 86$; hence*

$$a(86) > 2,000,000,000.$$

The additional exact values in Table 2 are also certified. For every

$$n \in \{86, 97\} \cup [107, 116] \cup [118, 128] \cup \{130\} \cup [132, 159] \cup [161, 254],$$

there is no base $b \leq 2,000,000,000$ with $C(b) = n$.

Proof. Theorem 4 shows that the production sweep increments exactly the counter $C(b)$ for every base in the finite interval. The exhaustive modulus frontier is $B - 1$, so no admissible $c < b \leq B$ is omitted. The production array stores $\min(C(b), 255)$, and for every target $1 \leq n \leq 254$ the equality $\min(C(b), 255) = n$ is equivalent to $C(b) = n$. Moreover, the production run recorded zero saturation events and maximum count 160, and the exact 32-bit validator confirmed every cell. Scanning the complete counter array therefore yields the stated first occurrences and exactly the 146 empty slots listed above. The independent recomputation and witness checks described in the preceding section reproduced every asserted row. $\square$

The canonical contiguous data file (the standard two-column OEIS term file, or *b-file*) therefore has exactly the rows $1 \leq n \leq 85$. The 23 rows of Table 2 remain separate so that gaps are never disguised as a contiguous b-file. Of the 254 inspected slots, 108 are populated—the 24 public terms and 84 new values—and the remaining 146 are exactly the undefined slots listed in Theorem 5.

Table 1: The 61 new consecutive terms of A255885.

n	$a(n)$	n	$a(n)$	n	$a(n)$
25	821339	46	12700801	67	7263026
26	762049	47	10285001	68	192099601
27	583201	48	25920001	69	258781249
28	902501	49	22619537	70	476280001
29	19601	50	11995201	71	644764049
30	470449	51	17322499	72	55989361
31	1822501	52	45158401	73	20353276
32	2073601	53	43468489	74	170067682
33	4120901	54	43560001	75	85377601
34	4064257	55	25401601	76	503004501
35	2822401	56	31360001	77	341510401
36	1587601	57	131584501	78	752716801
37	3484801	58	12966724	79	406425601
38	8294401	59	28755649	80	250905601
39	1555849	60	38102401	81	536869376
40	9331201	61	13231349	82	282240001
41	3650401	62	335923201	83	860606209
42	8193151	63	435974401	84	1111320001
43	4762801	64	180633601	85	1244678401
44	16769026	65	120963457		
45	18289153	66	146821249		

Table 2: Certified exact values beyond the consecutive endpoint. These rows are a support table, not an OEIS b-file.

n	$a(n)$	n	$a(n)$	n	$a(n)$
87	452838401	95	1866240001	103	1949771251
88	295293601	96	1124802449	104	881393644
89	146376476	98	457228801	105	514382401
90	189750626	99	416565249	106	553246849
91	524384524	100	1235661076	117	649296649
92	177902243	101	815264351	129	362074049
93	462003751	102	914636449	131	192119201
94	1296672301			160	768398401

7 A finite square-plus-one pattern

Some minima have a simple source of witnesses. If $b = m^2 + 1$ and a composite c divides m , then $c^2 \mid b - 1$ and hence $b^{c-1} \equiv 1 \pmod{c^2}$. Thus every composite divisor of m is counted by $C(b)$.

Among the 84 newly certified minima, 31 are of the form $m^2 + 1$. For six of them—at indices 32, 35, 38, 55, 79, and 85—the complete witness set is exactly the set of composite divisors of m . At the endpoint,

$$a(85) = 35280^2 + 1,$$

and the 85 witnesses are precisely the 85 composite divisors of 35280. The contrast with $a(29) = 140^2 + 1$ is instructive: eight witnesses are forced by composite divisors of 140, while 21 further moduli satisfy the congruence. These are exact observations about the certified finite data, not an asymptotic claim or a conjecture about future minima.

8 Conclusion

The inverse-modulus formulation turns the search for A255885 into a complete finite sieve. Prime-power root counts give the exact candidate space, while the interval-aware divisor projection controls the large- c regime without losing or duplicating solutions. The exhaustive production computation and independent recomputation extend the consecutive sequence from 24 to 85 terms, certify 23 further isolated indices, and prove the current finite endpoint statement $a(86) > 2 \cdot 10^9$. Determining $a(86)$ itself requires a new computation beyond the reviewed finite domain and the scope of this work.

Data availability

The public project repository and its synchronized Zenodo archive are listed in reference [5]. The release contains the frozen C17 production source and build instructions, the independent C17/Python validator, the contiguous and sparse result files, the witness certificate, validation reports, and the manuscript source and PDF. The exact values reported in this paper derive from the frozen campaign and validation artifacts distributed with that release.

AI use disclosure

Large language models were used as research and drafting assistants during the A255885 workflow, including OpenAI Codex and external LLM-based review systems from the ChatGPT, Claude, GitHub Copilot, DeepSeek, Gemini, Grok, Kimi, Meta, Qwen, MiniMax, and Z.ai GLM families. They are named at family level because exact model-version metadata were not consistently available in the local review artifacts.

Mathematical analysis. AI systems assisted with literature triage, checking proof structure, exploring finite certified data, and drafting exposition. Every mathematical claim retained here was checked against the cited sources, proved in the manuscript, or verified from the frozen exact data by the author.

Algorithm and code. AI systems assisted in generating and iteratively revising both the production and validation source code, as well as in identifying possible correctness, portability, overflow, persistence, and validation issues. The production and validator programs are separate codebases, but they share the broader AI-assisted mathematical framework and review environment;

their independence is therefore implementation-level, not process-level. The author selected and reconciled every change, specified the requirements, controlled every official run, and verified the frozen implementation and outputs.

Manuscript. OpenAI Codex assisted with the initial English draft, consistency checks, and LaTeX review. The author determined the claims, evidence boundaries, structure, and final wording.

Project workflow. AI tools did not autonomously authorize or operate the official computational campaign. Campaign execution, preservation of canonical artifacts, and the decision to report the results remained under the author’s control.

The scientific conclusions rely on the primary sources and frozen artifacts, including the result table, b-files, witness certificate, and independent validator, not on conversational assertions. The author assumes full scientific responsibility for all mathematical statements, the implementation, numerical results, validation boundaries, and conclusions, as well as for the citations and text of this article.

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